

MATH 3330 Homework 7

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Question 1

Show that if $h, h' : X \rightarrow Y$ are homotopic and $k, k' : Y \rightarrow Z$ are homotopic then $k \circ h$ and $k' \circ h'$ are homotopic.

Proof: Let $h, h' : X \rightarrow Y$ be homotopic and $k, k' : Y \rightarrow Z$ be homotopic. Since $h \simeq h'$, there exists some continuous map $F : X \times I \rightarrow Y$ with $F(x, 0) = h(x)$ and $F(x, 1) = h'(x)$. Similarly, because $k \simeq k'$, there exists some continuous map $G : Y \times I \rightarrow Z$ with $G(y, 0) = k(y)$ and $G(y, 1) = k'(y)$.

Now let $H(x, t) = G(F(x, t), t)$.

$$\begin{aligned} H(x, 0) &= G(F(x, 0), 0) = G(h(x), 0) = k(h(x)) = & k \circ h \\ H(x, 1) &= G(F(x, 1), 1) = G(h'(x), 1) = k'(h'(x)) = & k' \circ h' \end{aligned}$$

Since H is built from the composition of continuous functions, it must be continuous, so H is a homotopy and thus $k \circ h$ and $k' \circ h'$ are homotopic.

Question 2

Given spaces X and Y , the $[X, Y]$ denote the set of homotopy classes of maps of X into Y .

1. Let $I = [0, 1]$. Show that for any X , the set $[X, I]$ has a single element.
2. Show that if Y is path connected, the set $[I, Y]$ has a single element.

Proof: Let X and Y be spaces and $[X, Y]$ denote the set of homotopy classes of maps of X into Y . We must show that for $I = [0, 1]$, for any X , the set $[X, I]$ has a single element. Let $f, g : X \rightarrow I$ be homotopic. Define the homotopy $H(x, t) = (1 - t)f(x) + tg(x)$. H is well defined and continuous and has the boundaries $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$. Then $f \simeq g$ for any two maps into I so $[X, I]$ has only one element.

Proof: We must show that any two maps $f, g : I \rightarrow Y$ are homotopic. Since I is contractible, any map out of I is homotopic to a constant map. So, $H(s, t) = f((1 - t)s)$ is a homotopy from f to the constant map $c_{f(0)}$. Similarly $g \simeq c_{g(0)}$. Since Y is path connected, there exists a path $\delta : I \rightarrow Y$ from $f(0)$ to $g(0)$, which gives a homotopy from $c_{f(0)}$ to $c_{g(0)}$. By transitivity, $f \simeq c_{f(0)} \simeq c_{g(0)} \simeq g$, so $[I, Y]$ has a single element.

Question 3

A subset A of $\mathbb{R}^n$ is said to be **star convex** if for some point a_0 of A , all the line segments joining A to other points of A lie in A .

1. Find a star convex set that is not convex.
2. Show that if A is star convex, A is simply connected.

An example of a set that is star convex but not convex is

$A = [0, 2] \times [0, 1] \cup [1, 2] \times [1, 2] \in \mathbb{R}^2$. This forms an L-shaped region, so if you chose a point a_0 in the lower right corner (inside of $[1, 2] \times [0, 1]$) each line segment between a_0 and any other point in A lies entirely within A .

Proof: Let A be star convex, so for some $a_0 \in A$ each line segment connecting a_0 with b_0 for all $b_0 \in A$ lie entirely within A . We must show A is simply connected.

Let $x, y \in A$. Since A is star convex, there exists some $a_0 \in A$ that is connected to both x and y via some line segment. Then x and y must be connected by this intermediate point a_0 so A is path connected. Now let $f : [0, 1] \rightarrow X$ be a loop that starts at a point x_0 . Then we can define the following homotopy:

$$H(s, t) = (1 - t)f(s) + tx_0$$

So for each fixed s , the point $H(s, t)$ lies on the line segment between $f(s)$ and x_0 . This is a continuous homotopy. Since A is path connected and we can build a continuous homotopy, it must be simply connected.

Question 4

Let x_0 and x_1 be points of the path-connected space X . Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair α and β of paths from x_0 to x_1 , we have $\hat{\alpha} = \hat{\beta}$.

Hints

To show abelian, let f and g be loops. Let $x_1 = x_0$ and consider $\hat{f}(f)$ and $\hat{g}(f)$. To show $\hat{a} = \hat{b}$, use the isomorphism between $\pi_1(X, x_0)$ and $\pi_1(X, x_1)$ to get both are abelian and then compute $\hat{\alpha}(f)$ and $\hat{\alpha}(g)$. Use that $\beta\bar{\beta} = id$ and commutativity to get equality.

($\Rightarrow$) Suppose $\pi_1(X, x_0)$ is abelian. We must show that $\hat{\alpha} = \hat{\beta}$. Let α, β be paths from x_0 to x_1 and $[f] \in \pi_1(X, x_0)$. Then $\beta\bar{\alpha}$ is a loop at x_0 , so by commutativity:

$$\begin{aligned}
\hat{\alpha}([f]) &= [\bar{\alpha} * f * \alpha] \\
&= [\bar{\beta} * (\beta \bar{\alpha}) * f * \alpha] \\
&= [\bar{\beta} * f * (\beta \bar{\alpha}) * \alpha] \\
&= [\bar{\beta} * f * \beta] \\
&= \hat{\beta}([f])
\end{aligned}$$

so $\hat{\alpha} = \hat{\beta}$.

($\Leftarrow$) Suppose $\hat{\alpha} = \hat{\beta}$ for all paths α, β from x_0 to x_1 . Let f, g be loops at x_0 . Set $\alpha = e_{x_0}$ and $\beta = g$. Then:

$$\hat{\alpha}([f]) = [f], \quad \hat{\beta}([f]) = [\bar{g} * f * g]$$

Since $\hat{\alpha} = \hat{\beta}$, we get $[f] = [\bar{g} * f * g]$, hence $[f * g] = [g * f]$. Since f, g were arbitrary, $\pi_1(X, x_0)$ is abelian.

Question 5

Prove that if there are simply connected open subsets U and V of X such that $U \cup V = X$ and $U \cap V$ is nonempty and path connected, then X is simply connected.

Proof: Let there exist simply connected subsets $U, V \subseteq X$ such that $U \cup V = X$ and $U \cap V$ is nonempty and path connected. We must show that X is simply connected.

First, X is path connected so any two points $x, y \in X$ lie in U or V , each of which is path connected. Since $U \cap V$ is nonempty we can connect paths through the intersection. Thus X is path connected.

Pick a point $x_0 \in U \cap V$ and let $[f] \in \pi_1(X, x_0)$. We can subdivide I into subintervals so that f restricted to each subinterval lies entirely in U or V (Lebesgue number lemma). At each subdivision point $f(t_i)$, since $U \cap V$ is path connected, we can choose a path γ_i from x_0 to $f(t_i)$ lying in $U \cap V$. Inserting $\gamma_i * \bar{\gamma}_i$ at each subdivision point, we decompose $[f]$ into a product of loops each lying entirely in U or V . Since $\pi_1(U, x_0) = 0$ and $\pi_1(V, x_0) = 0$, each such loop is nullhomotopic, and therefore $[f]$ is trivial. Hence $\pi_1(X, x_0) = 0$, so X is simply connected.

Question 6 (optional)

Come up of an example where we can see that $\pi_1(X, x_0)$ is not abelian since $\hat{\alpha} \neq \hat{\beta}$ and also show that the isomorphisms between $\pi_1(X, x_0)$ and $\pi_1(X, x_1)$ depend on the path. (I'll work on this too...)

I wasn't able to come up with one myself, but I asked AI to see if it could find one and it did find one that works, being the figure eight $X = S^1 \vee S^1$:

Can you give me an example?

Come up of an example where we can see that $\pi_1(X, x_0)$ is not abelian since $\hat{\alpha} = \hat{\beta}$ and also show that the isomorphisms between $\pi_1(X, x_0)$ and $\pi_1(X, x_1)$ depend on the path. (I'll work on this too...)

A classic example is the **figure eight** $X = S^1 \vee S^1$, whose fundamental group is the free group $\pi_1(X, x_0) = F_2 = \langle a, b \rangle$, which is non-abelian.

Take x_0 to be the wedge point, and x_1 any other point on one of the circles. Let α and β be two different paths from x_0 to x_1 . Then:

$$\hat{\alpha}([f]) = [\bar{\alpha} * f * \alpha], \quad \hat{\beta}([f]) = [\bar{\beta} * f * \beta]$$

Since $ab \neq ba$ in F_2 , taking $f = a$ and $\gamma = \beta\bar{\alpha}$ a nontrivial loop, we get:

$$\hat{\alpha}([a]) = [\bar{\alpha} * a * \alpha] \neq [\bar{\beta} * a * \beta] = \hat{\beta}([a])$$

so $\hat{\alpha} \neq \hat{\beta}$, explicitly showing the isomorphism **depends on the choice of path**. This is exactly the failure of the abelian condition from Question 4: since $\pi_1(S^1 \vee S^1)$ is non-abelian, there must exist paths α, β with $\hat{\alpha} \neq \hat{\beta}$, and the figure eight gives a concrete place to see it.